

Computational graphs & the maths of AI

Lecture 3 – Three tools: linear algebra, calculus, and computation graphs

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Warm-up · three questions from last time

≈ 5 MIN · SHOUT IT OUT

1. A perceptron has $w = 2$ and $b = -1$. What is y when $x = 3$?
2. We stack three *linear* layers, with no activation functions. What family of functions can the stack represent — and why?
3. A perceptron takes three inputs. How many parameters does it have, counting the bias?

Warm-up · answers

1. $y = wx + b = (2 \times 3) - 1 = 6 - 1 = 5$.
2. **Still linear.** A composition of linear maps is a linear map, so three stacked linear layers collapse into one. Depth buys nothing until we add activation functions.
3. Three weights plus one bias: $3 + 1 = 4$ parameters.

| Part 1 · Linear algebra

Why we need this

A modern neural network operates on **tensors with millions of entries**. We need notation that scales — both for writing and for computing.

Every layer of a neural network is a *matmul* followed by a non-linear activation function applied element-wise.

- Matrices aren't that scary! Stacking survey into a data frame (n rows, k columns) gives you a matrix.
- We're largely dressing up familiar maths using new notation.

Scalars

The smallest unit. Just numbers.

$$1, 1.5, -0.001, \pi$$

Combine them with arithmetic to make expressions:

$$y = ((2 \times 3) + 5) \times 7 = 77$$

Or write the same thing as a **function**:

$$y = f(x) = (2x + 5) \times 7$$

A function is just a name for "input x , compute these steps, output y ."

Functions can be **composed**, i.e. $f(x) = g(h(x))$ where $h(x) = 2x + 5$ and $g(z) = 7z$.

Composition is the whole idea behind a deep network.

Vectors

A 1-D ordered list of scalars. Two we'll keep using:

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Addition / subtraction: element-wise, requires matching length.

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 1 + 4 \\ 2 + 5 \\ 3 + 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Scalar multiplication: multiply every entry.

$$3\mathbf{a} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

A survey-stats example

Rescaling everyone's height from cm to m is scalar multiplication of the height column

Row vs column — and the transpose

A vector can be a *column* $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ or a *row* $\begin{bmatrix} a_1 & a_2 \end{bmatrix}$.

The **transpose** ($'$ or T) swaps them:

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \iff \mathbf{a}^T = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$$



Note

The transpose is **only a re-orientation**. The numbers don't change. It exists so the operations later in this lecture line up dimension-wise.

Matrices

A 2-D array of scalars. Two indices — row and column.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Addition is again element-wise:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

A weight matrix in a neural network is just a 2-D table of numbers, ordered by “which output node” and “which input feature”.

Indexing a matrix

ROW FIRST, THEN COLUMN — ALWAYS

We write a_{ij} for the entry in **row** i , **column** j :

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

SINGLE ENTRIES

- $a_{12} = 2$
- $a_{31} = 7$
- $a_{23} = 6$

A whole row

Fix $i = 2$:

$$\begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

A whole column

Fix $j = 2$:

$$\begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}$$



$a_{12} = 2$ but $a_{21} = 4$ — **different entries**. Row always comes first, and this is the most common slip in the rest of this lecture.

TRY

What is a_{32} ? And what shape is column 3?

ANSWER

$a_{32} = 8$ — row 3, column 2. Column 3 is $\begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$, a 3×1 column vector.

Indexing in maths vs. indexing in code

Maths starts at 1

a_{11} is the **top-left** entry.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

So $a_{23} = 6$.

Python starts at 0

```
A[0, 0]    # 1    ← top-left
A[1, 2]    # 6    ← same entry as a_23
A[1]       # row 2: [4, 5, 6]
A[:, 1]    # column 2: [2, 5, 8]
```



Important

The **order is the same** — row, then column. Only the *starting number* differs: a_{ij} is `A[i-1, j-1]`.

You met this in yesterday's lab; you'll use it every day from here on.

Transposing a matrix

Swap rows and columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}, \quad \mathbf{A}^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}$$

TRY

What is $(\mathbf{A}^T)^T$?

ANSWER

A. Two transposes get you back where you started.

The dot product

THE MOST IMPORTANT OPERATION WE'LL SEE TODAY

For two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ (same length!):

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i$$

A **single number**. Sometimes written $\langle \mathbf{a}, \mathbf{b} \rangle$ or $\mathbf{a}^T \mathbf{b}$.

Worked example. Let $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$.

$$\mathbf{a} \cdot \mathbf{b} = (1)(4) + (2)(5) + (3)(6) = 4 + 10 + 18 = 32$$

Why it matters.

A neuron's pre-activation is $\sum_i w_i x_i + b$. That sum is a **dot product** $\mathbf{w} \cdot \mathbf{x}$.

- A weighted sum of survey responses is a dot product.
- A weighted average is a normalised dot product.

Matrix multiplication

Take $\mathbf{A} \in \mathbb{R}^{n \times k}$ and $\mathbf{B} \in \mathbb{R}^{k \times t}$.

The product $\mathbf{AB}$ is the $n \times t$ matrix whose entry (i, j) is the **dot product of row i of $\mathbf{A}$ and column j of $\mathbf{B}$** :

$$(\mathbf{AB})_{ij} = \sum_{\ell=1}^k A_{i\ell} B_{\ell j}$$

Dimension rule: the *inner* dimensions must match.

$$\underbrace{(n \times k)}_{\mathbf{A}} \cdot \underbrace{(k \times t)}_{\mathbf{B}} \longrightarrow (n \times t)$$

If they don't match, the product is *undefined*.

Matrix multiplication — worked example

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} \underbrace{1 \cdot 5 + 2 \cdot 7}_{=19} & \underbrace{1 \cdot 6 + 2 \cdot 8}_{=22} \\ \underbrace{3 \cdot 5 + 4 \cdot 7}_{=43} & \underbrace{3 \cdot 6 + 4 \cdot 8}_{=50} \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Two warnings.

- In general $\mathbf{AB} \neq \mathbf{BA}$. Order matters.
- Dimensions must match, i.e. if $\mathbf{A} \in \mathbb{R}^{3 \times 2}$, $\mathbf{B} \in \mathbb{R}^{2 \times 4} \rightarrow \mathbf{AB}$ is 3×4 . But $\mathbf{BA}$ is undefined.



$\mathbf{AB} \neq \mathbf{BA}$

Compute $\mathbf{BA}$ for the above matrices by hand. You should get a different number than $\mathbf{AB}$. This mismatch is why we need transposes to keep things orderly.

Why a layer is a matrix product

Yesterday: a hidden layer of three nodes, each computing $\sum_i w_i^{(h)} x_i + b^{(h)}$ for the same input $\mathbf{x}$.

Stack the per-node weights into a matrix $\mathbf{W} \in \mathbb{R}^{h \times k}$ and biases into a vector $\mathbf{b} \in \mathbb{R}^{1 \times h}$. The entire layer's pre-activation is:

$$\mathbf{x}\mathbf{W}^T + \mathbf{b} \in \mathbb{R}^{1 \times h}.$$

The transpose makes the inner dimensions match: $(1 \times k) \cdot (k \times h) \rightarrow (1 \times h)$.

I.e. One output per h nodes

Why a transpose? — dimensional intuition

Suppose $h = 2$ (two hidden nodes) and $k = 3$ (three input features).

- Weights for node 1: $\begin{bmatrix} w_1^{(1)} & w_2^{(1)} & w_3^{(1)} \end{bmatrix}$
- Weights for node 2: $\begin{bmatrix} w_1^{(2)} & w_2^{(2)} & w_3^{(2)} \end{bmatrix}$
- Stack as $\mathbf{W} = \begin{bmatrix} w_1^{(1)} & w_2^{(1)} & w_3^{(1)} \\ w_1^{(2)} & w_2^{(2)} & w_3^{(2)} \end{bmatrix}$ — shape (2×3) .

To compute $\mathbf{x} \cdot \mathbf{W}^T$ with $\mathbf{x}$ a row vector of length 3, we need $\mathbf{W}^T$ of shape (3×2) .

Put it together:

$$\mathbf{W}^T = \begin{bmatrix} w_1^{(1)} & w_1^{(2)} \\ w_2^{(1)} & w_2^{(2)} \\ w_3^{(1)} & w_3^{(2)} \end{bmatrix}$$

$$\text{For a given } \mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}, \quad \mathbf{x} \cdot \mathbf{W}^T = \begin{bmatrix} (x_1 w_1^{(1)} + x_2 w_2^{(1)} + x_3 w_3^{(1)}) & (x_1 w_1^{(2)} + x_2 w_2^{(2)} + x_3 w_3^{(2)}) \end{bmatrix}$$

So we get a two-length row vector output (one value per neuron).

From one observation to a batch

So far, one observation at a time. But, the labs feed many at once by stacking the observations as the **rows** of a matrix $\mathbf{X}$.

With 64 observations and our $h = 2, k = 3$ layer:

- $\mathbf{XW}^T$ is $(64 \times 3)(3 \times 2) \rightarrow (64 \times 2)$
- The same matrix product, just done simultaneously

Notice, we still need to add the biases $\mathbf{b}$...

- But $\mathbf{b}$ is (1×2) !?

Broadcasting — adding shapes that don't match

By the rules we wrote down earlier, the previous slide's addition is undefined.

Yet, if we run it in Python, it works anyway.

numpy and PyTorch use an extra rule called **broadcasting**:

- Line the two shapes up **from the right**
- Two dimensions are compatible when they are equal, or when one of them is 1 (or missing).
- The smaller array is then reused along the other's dimension

Suppose we have: $(3 \times 2) + (1 \times 2)$. Reusing along a dimension means copying the object until the shapes match, then adding as normal:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 10 & 20 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} + \underbrace{\begin{bmatrix} 10 & 20 \\ 10 & 20 \\ 10 & 20 \end{bmatrix}}_{\text{the re-use}} = \begin{bmatrix} 11 & 22 \\ 13 & 24 \\ 15 & 26 \end{bmatrix}$$

The middle version is ordinary same-shape addition — no new rules. Every row gets the same $\begin{bmatrix} 10 & 20 \end{bmatrix}$: one bias per node, reused for every observation. That is what $\mathbf{XW}^T + \mathbf{b}$ means in code.

Broadcasting the other way

Broadcasting can stretch a column too. Take $(3 \times 2) + (3 \times 1)$: from the right, 2 against 1 — so the 1 stretches across the columns; then 3 against 3 match.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} + \underbrace{\begin{bmatrix} 10 & 10 \\ 20 & 20 \\ 30 & 30 \end{bmatrix}}_{\text{stretching}} = \begin{bmatrix} 11 & 12 \\ 23 & 24 \\ 35 & 36 \end{bmatrix}$$

One number per row, reused along that row.

The direction of the stretch follows the **shape**.

- A (1×2) row spreads down the rows; a (3×1) column spreads across the columns.



Tip

You have broadcast before. Standardising a data frame subtracts a $(1 \times k)$ row of column means from an $(n \times k)$ matrix — the same pattern as the bias.

Broadcasting across dimensions

Later, when we stack matrices to get **3-D tensors**, we'll have more complicated shapes like $(64 \times 28 \times 28)$

Suppose we subtract something dimensionally smaller like (28×28) . Broadcasting works even here:

```
(64, 28, 28)
(28, 28)  → pad left with 1s → (1, 28, 28) → stretch → (64, 28, 28)
```

When one *shape* is shorter, **we pad it with 1s on the left**, then line up from the right as before.

- So the above shape is treated as $(1 \times 28 \times 28)$ and reused for all 64 square matrices.

E.g. $(2 \times 2 \times 2) + (2 \times 2)$:

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}}_{\text{a stack of two matrices}} + \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} = \begin{bmatrix} 11 & 22 \\ 33 & 44 \end{bmatrix}, \begin{bmatrix} 15 & 26 \\ 37 & 48 \end{bmatrix}$$

Each matrix in the stack gets the same (2×2) added — as if we had written $+ \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}, \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$, one copy per layer.



Warning

The padding is only ever on the **left**. $(64, 28, 28) + (64, 28)$ fails: from the right, 28 matches 28, then 28 hits 64. If you mean “one number per image” — say, dividing each image by its own maximum — the shape you want is $(64 \times 1 \times 1)$. Broadcasting follows the same rules for every element-wise operation, not only addition.

Try it — which of these broadcast?

TRY IT YOURSELF

For each pair of shapes, does the addition work — and if so, what shape comes out?

1. $(2 \times 3) + \text{a single number}$
2. $(64 \times 2) + (1 \times 2)$
3. $(64 \times 2) + (64,)$
4. $\begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix} + \begin{bmatrix} 1 & 2 \end{bmatrix}$ — that is, $(3 \times 1) + (1 \times 2)$

ANSWER

1. **Works** — a single number is compatible with every dimension. Shape (2×3) .
2. **Works** — from the right: 2 matches 2, then the 1 stretches over 64. Shape (64×2) . This is the layer bias.
3. **Error** — from the right: 2 against 64. A per-observation vector cannot be added this way. Reshape it to (64×1) and it broadcasts across the columns instead.
4. **Works** — *both* stretch. Written out: $\begin{bmatrix} 10 & 10 \\ 20 & 20 \\ 30 & 30 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 11 & 12 \\ 21 & 22 \\ 31 & 32 \end{bmatrix}$ — shape (3×2) , every pairwise sum.

The cost of a matrix product

We now know *what* a matrix product computes. How much work is it?

One entry of an $(n \times m)(m \times p)$ product is a dot product of length m : m **multiplications and $m - 1$ additions**. Each multiply or add is one **floating-point operation** (a **FLOP**), so one entry costs about $2m$ FLOPs.

The output has $n \times p$ entries:

$$\text{total cost} \approx n \times p \times 2m = 2nmp \text{ FLOPs}$$

Worked example. A layer reading a 28×28 pixel image (784 features) into 128 hidden nodes, for a batch of 64 images: $\mathbf{XW}^T$ is $(64 \times 784)(784 \times 128)$.

$$2 \times 64 \times 784 \times 128 = 12,845,056 \approx 13\text{M FLOPs}$$

... For *one* layer, of *one* forward pass.

Why GPUs?

The good news: none of that work has to happen in sequence.

Every entry of the output is independent of every other. Entry (i, j) needs row i of $\mathbf{X}$ and column j of $\mathbf{W}^T$.

- We can compute thousands of entries at the same time.

A **GPU** (graphics card) is built for exactly this because it calls the same easy computation, over thousands of points at once.

- A modern GPU manages on the order of 10^{14} FLOPs per second.
- At that rate, our 13M-FLOP layer takes roughly a ten-millionth of a second.

TRY IT YOURSELF

The output layer maps those 128 hidden values to 10 classes, same batch of 64. How many FLOPs?

ANSWER

$2 \times 64 \times 128 \times 10 = 163,840$ FLOPs — about 78 times cheaper than the $784 \rightarrow 128$ layer.

| Plenary · matrix products by hand

Plenary · matrix products by hand

≈ 10 MIN · PEN AND PAPER, IN PAIRS

FIVE PROBLEMS

For each, compute or say “undefined” (with a reason).

1. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = ?$

2. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \end{bmatrix} = ?$

3. $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = ?$

4. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} = ?$

5. $\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = ?$

Five minutes. Then we'll discuss together.

Plenary · answers

1. $\begin{bmatrix} 17 \\ 39 \end{bmatrix}$. A $(2 \times 2) \cdot (2 \times 1) \rightarrow (2 \times 1)$ matrix-vector product. Think of it as two dot products stacked.
2. $\begin{bmatrix} 7 & 8 \\ 9 & 10 \end{bmatrix}$. The first matrix is the **identity**. Multiplying by it leaves the second matrix unchanged.
3. 32. A row $\times$ column dot product. Single number.
4. $\begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix}$. A *column $\times$ row* — the outer product. Gives a matrix, not a scalar.
5. **Undefined**. Inner dimensions $2 \neq 3$ — can't do it.



Tip

The most common error in deep learning code is a shape mismatch. PyTorch raises a clear error when this happens. Train your eye now — and you'll save hours later.

Part 2 · Differentiation

What we want — and why

Tomorrow, for any parameter w in our model, we will want to know:

If I *nudge* w slightly, does the loss go up or down — and by how much?

This information is a **derivative**.



You'll have seen derivatives before.

We're going to re-introduce them with two restrictions:

- We don't care about proofs!
- We do care about computation — i.e. *getting a number out* given a function and a value of x .

The derivative — slope of a function

For $y = f(x)$, the derivative $\frac{dy}{dx}$ is the **slope** of f at point x .

For $y = ax^b + c$ (where a, b, c are scalars):

$$\frac{dy}{dx} = b \cdot a x^{b-1}$$

- Constants drop out (their derivative is zero).
- $b = 1$: $y = ax + c \Rightarrow \frac{dy}{dx} = a$.
- $b = 2$: $y = ax^2 + c \Rightarrow \frac{dy}{dx} = 2ax$.

Worked example.

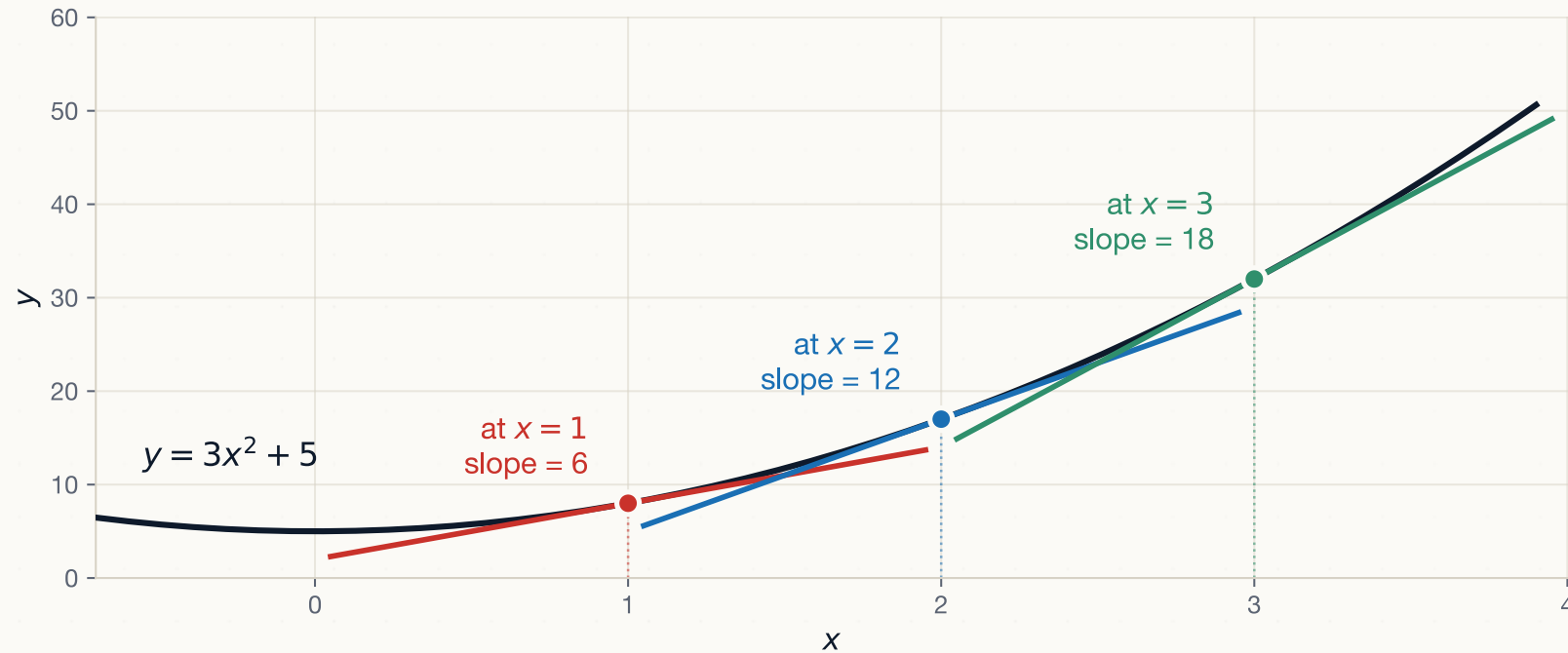
$$y = 3x^2 + 5.$$

$$\frac{dy}{dx} = 6x.$$

At $x = 1$: slope = 6. At $x = 2$: slope = 12. At $x = 0$: slope = 0 (local minimum!).

Seeing the slope

THE SAME WORKED EXAMPLE, DRAWN: $y = 3x^2 + 5$



The tangent is the straight line that just *touches* the curve at that point. Its steepness **is** the derivative there — so $\frac{dy}{dx} = 6x$ is a formula for the slope at **every** x , not one number. Steeper curve $\rightarrow$ bigger derivative.

A small cheat-sheet

Power rule

$$\frac{d}{dx} x^n = nx^{n-1}$$

Sum rule

$$\frac{d}{dx} (a + b) = \frac{da}{dx} + \frac{db}{dx}$$

Constant rule

$$\frac{d}{dx} c = 0$$

Exponential

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} e^{f(x)} = f'(x)e^{f(x)}$$

Natural log

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

Sigmoid (useful later)

$$\frac{d}{dx} \sigma(x) = \sigma(x)(1 - \sigma(x))$$



Note

You will not be asked to derive these. You **will** be asked to use them — both in this course and in the lab tomorrow.

A harder function

What about $y = (2x + 3)^2$?

You *could* expand to $y = 4x^2 + 12x + 9$ and apply the power rule: $\frac{dy}{dx} = 8x + 12$.

But what if expanding is messy — or impossible? We need a **mechanical** rule that doesn't depend on cleverness.

That rule is **the chain rule**.

The chain rule

For $y = f(g(x))$ — a function inside a function — define $u = g(x)$ so $y = f(u)$. Then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Worked example — $y = (2x + 3)^2$.

- Let $u = 2x + 3$. So $y = u^2$.
- $\frac{dy}{du} = 2u$ (power rule on u^2).
- $\frac{du}{dx} = 2$ (power rule on $2x + 3$).
- $\frac{dy}{dx} = 2u \cdot 2 = 4(2x + 3) = 8x + 12$. 

Same answer, mechanically.

The intuition. A small change in x propagates first through the inner function (u changes a little) and then through the outer function (y changes *because* u changed).

We want to know what the change through g *and* f is so we **multiply the two rates together**.

The chain rule, generalised

For $y = f_n(f_{n-1}(\cdots f_1(x) \cdots))$ — a chain of n functions:

$$\frac{dy}{dx} = \frac{dy}{du_{n-1}} \cdot \frac{du_{n-1}}{du_{n-2}} \cdots \frac{du_1}{dx}$$

Each piece is a small, easy derivative. Multiply them. Done.

This is the deepest reason deep learning works: any reasonable composition of operations can be differentiated **one operation at a time**, then assembled.

A longer chain

TAKE A FUNCTION WITH THREE NESTED OPERATIONS:

$$y = \left(\frac{1}{n}(a - x)^2\right)^{1/2}$$

Decompose: * $u_2 = a - x$ * $u_1 = u_2^2 = (a - x)^2$ * $y = (u_1/n)^{1/2}$

Apply the chain rule:

$$\frac{dy}{dx} = \underbrace{\frac{d}{du_1} \left[\left(\frac{u_1}{n} \right)^{1/2} \right]}_{\text{outer}} \cdot \underbrace{\frac{d}{du_2} [u_2^2]}_{\text{middle}} \cdot \underbrace{\frac{d}{dx} [a - x]}_{\text{inner}}$$

Three small derivatives. We don't even need to expand y to do this.

Partial derivatives & gradients

For a function of several variables, e.g. $y = f(a, b) = 3a^2 + 2b + 1$:

Compute **partial derivatives** — treat the other vars as constants:

$$\frac{\partial y}{\partial a} = 6a, \quad \frac{\partial y}{\partial b} = 2$$

Collect them into a **gradient vector**:

$$\nabla f = \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 6x_1 \\ 2 \end{bmatrix}$$

Why we care.

For a neural network with millions of weights, the loss is a function $L(\mathbf{W})$ from “all the weights” to “a single number”.

The gradient ∇L is a vector of the same size as $\mathbf{W}$ — one entry per weight. It tells us, for *each* weight, how the loss changes.

That is the thing we minimise tomorrow.

Checking a derivative numerically

Suppose a computer cannot differentiate a function. Can we approximate the derivative?

The derivative asks how much would the output change if we adjust an input by a tiny interval.

So, empirically, we can take a small step ε either side of x and measure that change:

$$f'(x) \approx \frac{f(x + \varepsilon) - f(x - \varepsilon)}{2\varepsilon}$$

This is called the **central difference**. It is the empirical evaluation of a gradient *at a given point*.

Worked example. Suppose $f(x) = x^2$, and we evaluate at $x = 3$.

With $\varepsilon = 0.01$:

$$\frac{f(3.01) - f(2.99)}{0.02} = \frac{9.0601 - 8.9401}{0.02} = \frac{0.1200}{0.02} = 6.0000$$

Formally, we know $f'(x) = 2x$, so $f'(3) = 6$ – they match!

Try it on the sigmoid

The cheat-sheet claims $\sigma'(x) = \sigma(x)(1 - \sigma(x))$. We took that on trust. Now we can check it.

TRY IT YOURSELF

Estimate $\sigma'(0)$ with the central difference and $\varepsilon = 0.01$.

To 4 d.p.: $\sigma(0.01) = 0.5025$ and $\sigma(-0.01) = 0.4975$.

ANSWER

$$\sigma'(0) \approx \frac{0.5025 - 0.4975}{0.02} = \frac{0.0050}{0.02} = 0.25$$

The rule agrees: $\sigma(0) = \frac{1}{1+e^0} = 0.5$, so $\sigma'(0) = 0.5 \times 0.5 = 0.25$.

Why don't we just use the central difference?

The central difference has one flaw that rules it out for training, but is useful for testing calculations.

It's far too slow

- Checking one parameter means nudging *one* w and rerunning the whole network – so, one forward pass per parameter.
- A model with millions of parameters would need millions of forward passes for a *single* training step.

Why it's still useful

- Autograd libraries use central difference to test themselves
 - Compute each derivative exactly
 - Compute the central difference
 - Only allow the algorithm to pass if they agree (called **gradient checking**)
- If our chain-rule code is wrong, gradient checking flags it



We can calculate entire network gradients with 1 forward and 1 backward pass

We'll see this tomorrow!

| **Activity · chain-rule exercises**

Activity · chain rule

≈ 8 MIN · PEN AND PAPER

For each y , find $\frac{dy}{dx}$ — using the chain rule, *without* expanding.

1. $y = (3x + 1)^4$

2. $y = e^{2x+5}$

3. $y = \ln(x^2 + 1)$

4. $y = \sigma(3x)$ where $\sigma(z) = \frac{1}{1+e^{-z}}$

5. $y = (a - x)^2$ — find $\frac{dy}{dx}$, treating a as a constant.

Activity · answers

1. $\frac{dy}{dx} = 4(3x + 1)^3 \cdot 3 = 12(3x + 1)^3.$

2. $\frac{dy}{dx} = e^{2x+5} \cdot 2 = 2e^{2x+5}.$

3. $\frac{dy}{dx} = \frac{1}{x^2 + 1} \cdot 2x = \frac{2x}{x^2 + 1}.$

4. $\frac{dy}{dx} = \sigma(3x)(1 - \sigma(3x)) \cdot 3.$

5. $\frac{dy}{dx} = 2(a - x) \cdot (-1) = -2(a - x).$

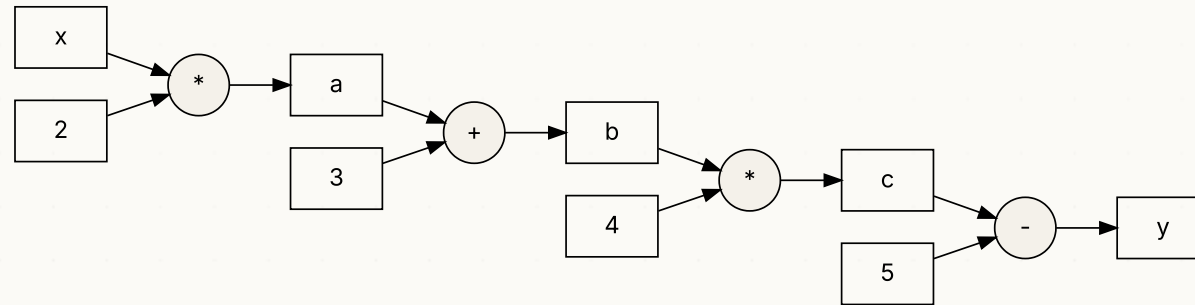
| Part 3 · Computational graphs

A function as a picture

Take $f(x) = 4(2x + 3) - 5$. It has:

- **Values:** $x, 2, 3, 4, 5$.
- **Operations:** $\times, +, -$.
- **Order** (BIDMAS).

Draw it as a graph — left-to-right, with operations as circles and values as boxes:



A **computational graph** is just a flow chart for a calculation. Each node either holds a value or performs an operation. Edges show the order of operations.

The pieces — value nodes and operation nodes

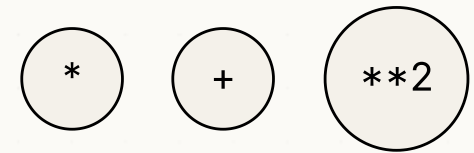
Value nodes

- Contain **data** — scalars, vectors, matrices, or higher-dimensional tensors.
- Will later also hold **gradients** (their slot for $\frac{\partial L}{\partial}$).
- Drawn as boxes.

x	data = 0.0	grad = 0.0
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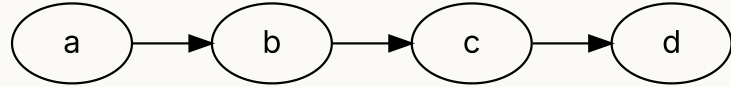
OPERATION NODES

- Apply a **function** to incoming values.
- Arrows in = arguments. Arrow out = result.
- Drawn as circles.

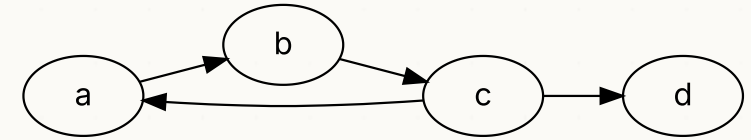


Acyclic — almost always

We use **directed acyclic graphs** (DAGs) — no operation's output ever flows back to an *earlier* node.



Acyclic — what we use



Cyclic — pathological for plain neural nets



Note

We **will** see cycles later — recurrent networks on Day 9, where the same operation is reused across time steps. But each unrolled time-step is still a DAG.

Graphs are fairly common in social sciences

SOCIAL-SCIENCE GRAPHS

A computational graph is structurally identical to:

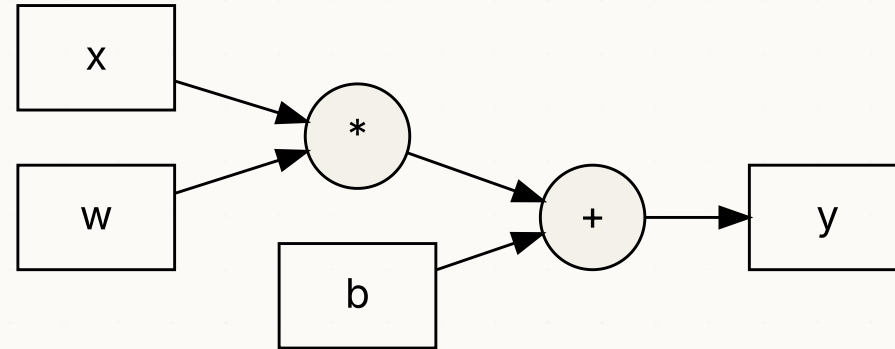
- A **causal DAG** in causal inference (cf. Judea Pearl's work).
- A **factor graph** in probabilistic modelling.
- A **dataflow diagram** in software engineering.
- A **path diagram** in structural equation models.

The shared idea: a calculation is a *structure of operations on values*.

The structure itself carries information, by telling us what depends on what, what we can compute first, and what to differentiate with respect to what.

The perceptron as a graph

Yesterday's linear perceptron $y = wx + b$ becomes:



(With an activation function σ added between the $+$ and the output, you'd insert one more node — same picture, one more step.)

Every node has a local derivative we can compute.

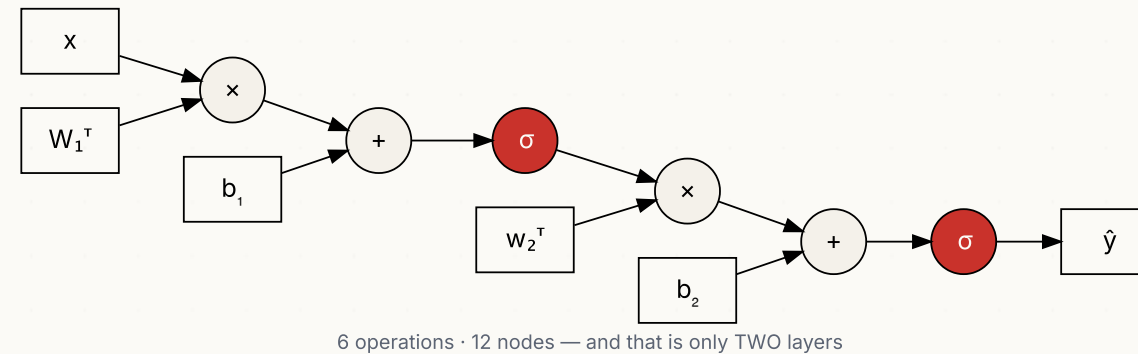
Using the cheat-sheet from earlier, we can assign the form of these derivatives when we build the graph, to save compute.

A network as a graph

Yesterday's two-layer network was:

$$\hat{y} = \sigma \left(\sigma \left(\mathbf{x} \mathbf{W}^{(1)\top} + \mathbf{b}^{(1)} \right) \mathbf{w}^{(2)\top} + b^{(2)} \right)$$

BUILD IT ONE OPERATION AT A TIME — WATCH THE NODE COUNT



We **multiply**, **add**, “**squash**”, twice. What grows is the *tracking*: three operations per layer, and every one needs its own local derivative.

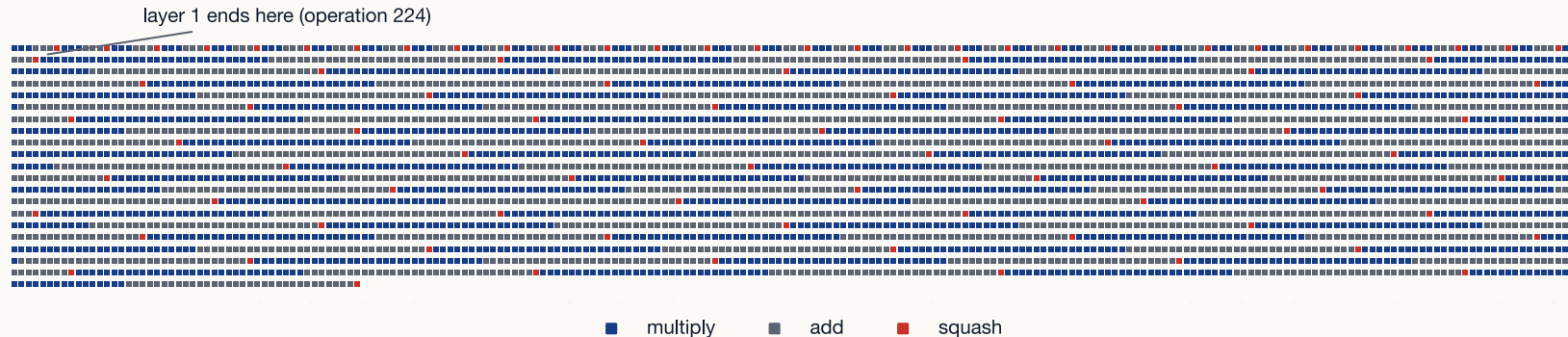
In **Labs 3–4** you build exactly this, where every single arithmetic operation becomes a node in your **Value** graph. A tiny network makes a surprisingly big graph — and that is precisely why we let the computer walk it for us.

Scaling up — three hidden layers of 32

This is still a *tiny* deep learning model:

- 3 inputs,
- 3 hidden layers of 32 nodes
- One output node

To make it fit, one dot represents a multiply, add, or squash, in running order.



For **one prediction**:

- Multiplies: $96 + 1,024 + 1,024 + 32 = 2,176$
- Adds: 2,176 (97 of them bias adds)
- Squashes: 97 — one per neuron

Grouped into matrix operations, this whole ribbon is still $4 \times$ (multiply, add, squash) = 12 nodes. The dots are what the computer does underneath.

For context, a forward pass of GPT-3 requires

How does the computer know what to run first?

The graph itself is not numbered.

- Instead, the order is extracted from the **arrows**

1. A node is **ready** when every arrow *into* it comes from a node that has already run.
2. Every ready node is added to a *heap*
3. Run any node in the heap then mark it done.
4. Repeat until every node has run.

A list built this way is called a **topological order** of the graph.

Two things worth noticing:

- The order is **not unique**. Layer 1 can run the multiplications in any order
 - Layer 2 multiples rely on layer-1 finishing
- The heap only clears because the graph is *acyclic*.
 - When cycles exist, (at least) two nodes are waiting for each other

The two passes — preview of tomorrow

Forward pass

- Walk the graph **left-to-right**.
- At each node, compute its output using the operation and incoming values.
- Store the output (we'll need it).
- End: a prediction.

BACKWARD PASS

- Walk the graph **right-to-left**.
- At each node, compute $\frac{\partial L}{\partial(\text{this node})}$ using the chain rule and local derivatives.
- End: a gradient for every parameter.



Tip

This is what frameworks (PyTorch, JAX, TensorFlow) automate. Tomorrow you'll write a tiny version of it yourself, in Python.

Take stock

WHERE YOU ARE AT THE END OF DAY 3

You can now:

1. Multiply matrices and check shapes by hand.
2. Differentiate simple expressions, including with the chain rule.
3. Draw any expression, from a simple function to a multi-layer neural network, as a directed acyclic graph.

What we don't have:

- A **way to put loss into the graph and let gradients flow backwards.**



Tip

You have all three tools. Tomorrow we put them together.

| Lab brief

Tomorrow morning's lab — 90 minutes

GOALS

In Python, build a small **computational graph library** from scratch.

You will implement:

1. A `Value` class that stores `data`, `grad`, and a list of "child" values.
2. Methods `__add__`, `__mul__`, and `__pow__` that build a graph as you do arithmetic.
3. A small test: write `c = a * b + 1` for `Value`s `a`, `b`, and verify the graph has the right structure.

Deliverable. A working `Value` class. We will add gradient computation to it in the following lab, which turns this library into a fully-working autodiff system.

This comes from Andrej Karpathy's `micrograd` scripts.

See you tomorrow

LECTURE 4 — BACKPROPAGATION

If you want a head start: skim Karpathy's *micrograd* GitHub README tonight. No need to read the implementation yet as we'll write it together tomorrow.